What the Coherence Volume Was

An RG Reduction of CHI, and a Self-Certifying Scaling Estimator

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Abstract

The CTMT "coherence volume" $\chi = Mv^2/(\Phi gh\rho)$ worked in practice: a single calibration anchor, tuned on one wind turbine, transported to others, and the same trick gave usable automotive fuel estimates. This note asks the elimination question of it—what *was* χ, in the mature Resolution-Geometry vocabulary?—and answers cleanly. Writing $Y = k\chi$ and taking logarithms, χ is a monomial with fixed exponents (a Buckingham–Pi group), and χ is exactly the coordinate along the single *resolved direction* of the log-observation geometry: the exponent covector $a = \partial \log Y/\partial \log \theta$ is that direction (row space of the log-Jacobian, rank one), the iso-χ surfaces are the null sector, and the calibration constant k is the anchor that fixes the offset. Single-anchor transport works precisely when that resolved direction is *index-locked* (rank-stable) and *flat* (holonomy-trivial): one frame, one k. The "coherence class" is that flat chart; the "Fisher seal" is the index lock; and the "seepage" failure the original paper feared is exactly the resolved direction drifting/rotating with regime—a rank/holonomy event. RG turns this from an assertion into an operational test. We ship the resulting estimator: it *learns* the exponents from data (recovering hand-picked CHI as a special case), certifies index-lock/flatness, and predicts from one anchor—flagging when a point has left the validated class. On a clean turbine it recovers $P \propto \rho A v^3$ (exponents $1.04, 0.99, 2.99$; drift 3%) and transports at a few-percent error; on a pitch-capped turbine it *flags* the regime break (v-exponent $2.99 \to 0.27$, drift 84%) that would silently break a fixed k; on fuel it correctly reports a band-limited exponent drifting $1.7 \to 2.5$ (rolling to aero). It is dimensional power-law scaling with a validity certificate—fast, scalable, and honest about its own range.

1 What worked, and the question

The coherence volume $\chi = Mv^2/(\Phi gh\rho)$ was an early CTMT object, and it earned its keep: calibrated once on a turbine it predicted others; it recovered Betz v^3 scaling after $M = \rho Av$; it gave plausible fuel-consumption transport from a single anchor. The mature program does not leave such a thing mystified. It asks: assuming a reader who knows all of standard mathematics, *what is* χ, and *why did single-anchor transport work when it did and fail when it did*?

2 The reduction

Write the working relation $Y = k\chi$ and take logarithms:

$$\log Y = \log k + \underbrace{1 \cdot \log M + 2 \cdot \log v - \log \Phi - \log h - \log \rho}_{\log \chi} = \log k + a \cdot \log \theta,$$

with $\theta = (M, v, \Phi, h, \rho)$ and the *fixed* exponent covector $a = (1, 2, -1, -1, -1)$. Two classical facts are now visible: χ is a *Buckingham–Pi monomial* (dimensionally closed, hence the m^3 units), and the forward map $\log \theta \mapsto \log Y$ is *affine with constant gradient a*.

Claim 1 (RG identity of χ)**.** χ is the coordinate along the single resolved direction of the log-observation geometry. Concretely, with observation map $Y(\theta)$ and log-Jacobian $J = \partial \log Y / \partial \log \theta = a$:

CHI object	RG object	established primitive
χ (coherence volume)	resolved coordinate	value of a rank-1 log-monomial
exponent vector a	resolved direction (Im J)	Pi-group exponents / cokernel row
iso-χ surfaces	null sector (ker J)	level sets / gauge orbit
calibration k	anchor (offset along Im J)	prefactor of a power law
"coherence class"	index-locked, flat chart	region of constant rank J, no holonomy
"Fisher seal"	index lock	rank-stability of Fisher $= J^\top \Sigma^{-1} J$
"seepage" (regime leak)	resolved-direction drift	rank change / nontrivial holonomy
volatility $\Xi = \frac{d}{dt} \log \chi$	drift along resolved coord	log-derivative
hazard $\lambda_{\min}/\lambda_{\max}$	distance to a wall	conditioning / stratum boundary

Claim 2 (Why single-anchor transport works exactly when it does)**.** Given a *fixed* resolved direction a, one anchor (θ_0, Y_0) fixes the offset $\log k = \log Y_0 - a \cdot \log \theta_0$ and determines Y everywhere. This is valid iff a is constant across the operating range—i.e. iff the resolved sector is index-locked (rank-stable) and flat (holonomy-trivial). If a drifts or rotates with regime (a wall / monodromy event), a single k cannot transport. This is precisely the "seepage" the original paper diagnosed informally; RG names it and makes it measurable.

Remark 1 (This is deflationary, and that is the point). The reduction says χ is not new: it is dimensional power-law scaling, and its transportability is governed by rank-stability of the log-Jacobian direction. That is exactly the elimination ethos—every RG object reduces to an established one. The value is not a new invariant but the *operational* completion: RG converts the vague "coherence class" into a checkable index-lock/flatness certificate, and generalizes the hand-picked monomial to a learned one.

3 The estimator

The companion module `RG_CHI_estimator.py` implements the reading directly:

1. **Learn the resolved direction.** Regress $\log Y$ on $\log \theta$ to obtain the exponents a and offset $\log k$ (forcing a to the classic values recovers hand-picked CHI). Cost $O(nd^2)$; pure linear algebra.

2. **Certify (index lock / flatness).** Refit a on windows across the operating range and measure the full-vector exponent drift plus the log-linear R^2. Small drift and high $R^2 \Rightarrow$ single-anchor transport is valid across the class; otherwise the tool *flags* a regime change or a band limit and reports the per-window exponents.

3. **Anchor and predict.** Fix k from one point; predict $Y = k \prod \theta_i^{a_i}$ with a multiplicative interval from the log-residual scatter.

4. **Guard.** `in_class` tests whether a new point lies inside the validated resolved-coordinate range; `volatility` returns $\Xi = \frac{d}{dt} \log \chi$ for early warning.

system	learned exponents	certificate	verdict
turbine (clean $\rho A v^3$)	1.04, 0.99, 2.99	R^2=.998, drift 3%	PASS: one k transports ($\sim$2%)
pitch-capped turbine	v: 2.99$\to$0.27	R^2=.89, drift 84%	FLAG: regime break caught
fuel (rolling+aero)	v: 1.7$\to$2.5	R^2=.99, drift 22%	FLAG: band-limited, exponent drifts

The clean case reproduces CHI's success and *explains* it (flat resolved direction). The pitch-capped case reproduces CHI's feared failure and *catches* it before it mis-predicts. The fuel case is the honest correction: fuel is not a single monomial, so a single k is valid only within a speed band, and the tool says so and gives the local exponent rather than a false global constant.

4 Non-claims

Non-claim 1. No physics. χ is a dimensional scaling coordinate; the estimator is log-linear regression plus a rank-stability certificate. Nothing here asserts a law of nature.

Non-claim 2. No novelty of primitive. χ reduces to a Buckingham–Pi monomial; the transport criterion reduces to Fisher/Jacobian rank-stability (the index lock). The contribution is the operational certificate and the data-driven generalization, not a new invariant.

Non-claim 3. The estimator is a first-order scaling and diagnostic tool, not a replacement for CFD/FEM or detailed models where fine-scale structure must be resolved.

5 Conclusion

The coherence volume was an intuition-driven rediscovery of dimensional power-law scaling, and its single-anchor transport was the statement—unnamed at the time—that the resolved direction of the log-observation geometry is index-locked and flat over a regime. RG supplies the missing half: a certificate that decides, from data, whether that condition holds, and a learned generalization that no longer depends on guessing the monomial. What was an early, naive formula becomes a small, fast, self-certifying estimator that predicts where it is valid and refuses where it is not.

References

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